

A Compact Hybrid ResNet-ViT for Automatic Cancer Detection in Histopathological Images

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Abstract

Cancer diagnosis delays represent a critical public health uphill battle, particularly in healthcare systems under strain of professional burnout and bureaucratic inefficiencies. Computer-Aided-Diagnosis (CAD) offer a compelling pathway to alleviate this burden by automating the analysis of histopathological images. This work proposes a lightweight hybrid ResNet-ViT architecture that unifies both paradigms: integrating a residual convolutional branch for local texture extraction with a patch-based transformer branch for global context reasoning. Feature integration is done through a Bi-Directional Cross Guidance (BDCG) fusion mechanism while classification is handled by a Prototype-Anchored Similarity Head (PASH) based on learned class prototypes, both adapted from earlier literature. Three preprocessing and training configurations were effectively evaluated on PatchCamelyon dataset, including normalization, data augmentation and CLAHE enhancement. The final model achieved a ROC-AUC of 0.9791 with limited hardware resources, demonstrating that competitive performance is viable without large-scale backbones and architectures. These results support the practicality of compact hybrid architectural as tools for resource-limited CAD deployment. The code of the project is available at <https://github.com/javixfdez/Compact-Hybrid-ResNet-ViT>.

1. Introduction

Cancer remains a major public health problem and one of the leading causes of death worldwide [1]. Countries such as Spain face an increasing crisis regarding the slowness and delay of cancer diagnoses. This vulnerability was highlighted during the recent pandemic, which severely disrupted the healthcare infrastructure. For instance, diagnoses for critical illnesses like breast and prostate cancers have shifted toward advanced and life-threatening stages [2].

A massive contributor to this problem is the severe strain that is placed on medical professionals; causing an alarmingly 24% of prevalence of burnout syndrome among Spanish doctors [3]. This hurdle is fueled by an inefficient system and an overwhelming bureaucratic workload which consumes time and energy.

The consequences of this systematic issue end up in the hands of patients and their families. Clinically, delayed diagnoses are correlated with decreased survival rates and force patients to undergo aggressive and invasive treatments. On the other hand, navigating this slow and uncoordinated process leaves patients emotionally devastated, generating severe psychological distress including anxiety and depression [4].

To alleviate this disorganized and inefficient diagnostic pathway, the integrating of automated solutions like Computer-

Aided-Diagnosis (CAD) and Computational Pathology (CPath) has become an imperative necessity as demonstrated in [5][6].

1.1. From Machine Learning to Visual Transformers

Historically, the automated detection and classification of medical images relied heavily on traditional Machine Learning algorithms, such as Support Vector Machines (SVM), k-Nearest-Neighbors (KNN) or Random Forests [7]. While providing a foundational step, these models struggled to scale and adapt to the high-dimensional, complex nature of histopathological data, leading to a plateau in their diagnostic performance [8]. To overcome these roadblocks, the field experienced a shift toward Deep Learning (DL), specifically leveraging Convolutional Neural Networks (CNNs). For instance, architectures like VGG16 or DenseNet121 achieved impressive accuracies in the classification of colorectal tissue as proven in [9].

Among these architectures, new approaches have emerged such as the Residual Networks (ResNet), introducing the residual connections that enabled information to bypass layers, facilitating gradient flow and preventing the vanishing gradient problem [6]. Making ResNet highly robust for capturing localized textures and fine-grained details in histopathological tasks.

Despite the widespread success of CNNs and ResNets, they possess an architectural limitation: the locality bias. These algorithms rely on localized convolutional filters which have restricted receptive fields that limit their ability to model global spatial relationships and dependencies across images.

To bridge this gap, Vision Transformers (ViTs), inspired by their massive success in Natural Language Processing, were finally introduced in the computer vision domain [10]. ViTs process images as a set of patches and are based on self-attention mechanisms. These attention mechanisms acts as a dynamic selection process that computes attention weights, allowing the model to focus on most diagnostically relevant regions of the image while effectively suppressing irrelevant background noise. Due to this, ViTs excel at recognizing broader structural asymmetries and heterogeneous tissue architectures as evidenced in [11].

While CNNs, ResNets and ViTs have shown immense potential in the CAD world, deploying the architectures in isolation limits the diagnostic capability. Standalone CNNs excel at extracting fine-grained, local features like textures and borders but they lack architectural context and struggle to model spatial relations. Conversely, ViTs capture successfully large range dependencies and global relationships yet strive at extracting lower level details [12].

In this work we solve these existing limitations by introducing a hybrid ResNet-ViT architecture, which synthetically fuses the inductive locality bias of CNNs with the global reasoning of

Transformers. The remaining part of the paper is organized as follows: in section II the dataset and model architecture are described, in section III, the results and discussion are presented, finally, we provide the conclusions in section IV.

2. Methods

All experiments and procedures detailed in this study were conducted within the cloud-based Kaggle Notebook, restricted to 30GB CPU RAM and 12-hour session timeout. Training was accelerated using a synchronous distributed strategy across two NVIDIA Tesla T4 GPUs (16GB VRAM each).

2.1. Dataset

The proposed models are trained and evaluated using the *Histopathologic Cancer Detection* dataset, accessible via Kaggle [13] and derived from the *PatchCameylon* (PCam) [14]. It comprises 220025 training and 57458 test images of patches extracted from lymph node sections with size 96x96 pixels on three channels (RGB). A representative sample is provided in Figure 2.

2.2. Image pre-processing and Data Pipeline

The initial pipeline performed minimal preprocessing where images were decoded and normalized to [0,1] [7], feeding a single-GPU training loop. This configuration served as the baseline, kept simple to isolate model behavior from preprocessing effects.

As training was scaled to a dual-GPU setup, the pipeline was adapted to a distributed strategy, doubling the batch size from 64 to 128 to maintain both devices occupied. This stage also introduced a more robust TIFF decoder, as histopathological images can exhibit diverse compression formats that simple readers handle inconsistently. Building on this distributed setup, data augmentation was incorporated consisting of random horizontal and vertical flips combined with rotations, exploiting the orientation-agnostic nature of histopathological images to increase training diversity.

The final configuration introduced Contrast Limited Adaptive Histogram Equalization (CLAHE) as an explicit preprocessing step. Histopathological slides are particularly susceptible to staining and contrast variability [8]. CLAHE addresses this by enhancing local contrast within regions independently, normalizing the appearance of tissue structures without amplifying noise, a recurrent limitation of global histogram equalization methods [15].

2.3. Models Architecture

The baseline was designed using a lightweight custom ResNet. Starting from this baseline, we further added additional modifications as explained in section 2.2. The network follows the residual learning paradigm consisting of a stem convolution followed by two residual blocks and global average pooling classifier, which effectively mitigate the vanishing gradient problem, as evidenced in [6]. However, this architecture exposes two fundamental limitations: the shallow residual stack lacks the capacity to capture tissue features across multiple spatial scales simultaneously, and the inherently local nature of convolutional models prevents reasoning about long-range spatial relationships [11]. This configuration is referred to as **AR1** throughout this work.

Hybrid ResNet-ViT model addresses the limitations of AR1 by combining a convolutional branch for local feature extraction with a lightweight Vision Transformer branch for contextual reasoning, unifying both through two components: a Bi-Directional Cross-Guidance fusion mechanism and a Prototype-Anchored Similarity Head classifier. The proposed architecture is a compact and lightweight variation of the one proposed in [8].

The local branch follows a residual architecture, processing the input image through a stem convolution and two residual blocks to produce a spatial feature map. This map is then flattened into a sequence of spatial tokens, preserving the positional structure of local texture and edge features. The global branch processes the same input independently using a patch embedding strategy which partitions the image into non-overlapping patches and projects each into a shared embedding space, further explained in [10]. Positional encodings are added to retain spatial ordering. A single Transformer encoder block, constrained by hardware and memory limitations, comprising multi-head self-attention [16] and an expansion-compression MLP allows the model to capture long-range dependencies, encoding the anatomical context that local convolutions cannot access, as [17] stated.

The two branches are reconciled by a Bi-Directional Cross-Guidance (BDCG) fusion mechanism, adapted from cross-attention fusion approaches in multi-modal learning [8]. Rather than concatenating or averaging the two feature sequences, BDCG enables bidirectional interaction: the ViT branch uses global context to guide the ResNet branch on which local textures are relevant, while the ResNet branch constrains the ViT branch from making global assumptions that contradict local evidence. The pooled outputs of both refined sequences are concatenated into a unified representation and passed to a Prototype-Anchored Similarity Head (PASH) classifier. Instead of learning a linear decision boundary, PASH maintains two learnable prototype vectors, one per class, and classifies each input according to its similarity to these anchors via squared Euclidean distances transformed into class probabilities through a Student's t-distribution kernel, improving resilience to clinical outliers compared to standard Softmax or Sigmoid classifiers [18], representation of this backbone in Figure 1.

This hybrid model was evaluated under two preprocessing configurations, referred to as **AR2** (with data augmentation) and **AR3** (with CLAHE preprocessing), as detailed in Subsection 2.2.

3. Experiments and Discussion

3.1. Models Configurations

All experiments shared a common foundation: the Adam optimizer with binary cross-entropy loss, EarlyStopping monitoring validation AUC, and a fixed random seed of 42. **AR1** was trained on a single GPU for 8 epochs with a batch size of 32, an initial learning rate of 10^{-3} , and ReduceLROnPlateau as the only scheduling mechanism. A 5-fold Stratified Cross-Validation strategy was employed to ensure comprehensive performance estimation across the entire dataset.

AR2 introduced `MirroredStrategy` across two GPUs, scaling the global batch size to 128 (64 per device), and incorporated GPU-accelerated data augmentation consisting of random horizontal and vertical flips and rotations up to 20° . Training was extended to 60 epochs with a cap of 1,000 steps per epoch. Due to the quadratic memory complexity $\mathcal{O}(N^2)$ intro-

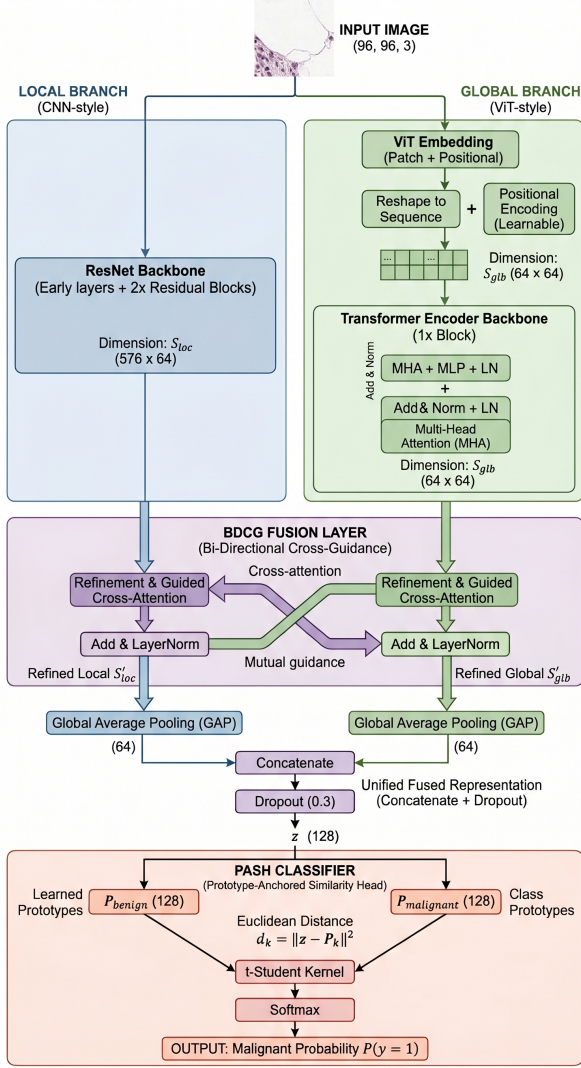


Figure 1: *Hybrid Backbone, common for AR2 and AR3 architectures, excluding preprocessing.*

duced by the Multi-Head Attention mechanisms [16] within the hybrid architecture, training across multiple folds exceeded the hardware and temporal constraints of the Kaggle cloud environment. Consequently, AR2 and AR3 were evaluated using a single holdout split (80% training, 20% validation), justified by the scale of the dataset, which provides thousands of unique validation instances sufficient to reliably estimate generalization.

AR3 replaced augmentation with CLAHE preprocessing and introduced a cosine annealing schedule with a 3 epoch linear warm up, replacing the reactive plateau-based approach. The warm-up was necessary to stabilize the early gradient dynamics of the BDCG attention layers and the randomly initialized PASH prototypes. The base learning rate was set to 10^{-4} , decaying smoothly to 10^{-6} over 40 epochs.

3.2. Models Performances

Table 1 summarizes the ROC-AUC and accuracy at a 0.5 decision threshold for all configurations evaluated on the held-out validation set. AR1 obtained a ROC-AUC of 0.9738 and an

Table 1: *Performance comparison of the evaluated architectures (in bold best result).*

Model Architecture	ROC-AUC	Accuracy (Threshold=0.5)
ResNet Baseline	0.9738	0.9100
ResNet-ViT (Old Pipeline)	0.9791	0.9318
ResNet-ViT (Robust Pipeline)	0.9418	0.8506
ResNet-ViT (CLAHE)	0.9517	0.8518

accuracy of 91%, reinforcing the long-standing effectiveness of residual architectures in CAD tasks as manifested in [14]. AR2 produced the highest accuracy across all configurations (93.18%) and improved ROC-AUC to 0.9791, demonstrating that the BDCG fusion and PASH classifier provide meaningful discriminative gains over the purely convolutional baseline when preprocessing conditions are held constant.

AR3 shows a divergence between ROC-AUC and accuracy, achieving a ROC-AUC of 0.9517 with an accuracy of approximately 85%. This is attributable to the increased diversity of training conditions introduced by CLAHE, which broadens the learned feature distribution at the cost of sharpness at the default decision threshold. Importantly, in CAD applications, ROC-AUC offers a more dependable assessment of a model’s discriminative capacity; the reduction in accuracy primarily reflects a threshold calibration issue rather than a genuine loss in class separability.

Qualitative inspection of the model predictions in Figure 2 further supports these results. High-confidence scores above 99% are assigned to malignant patches, while benign tissue is correctly classified with near-zero malignancy probability.

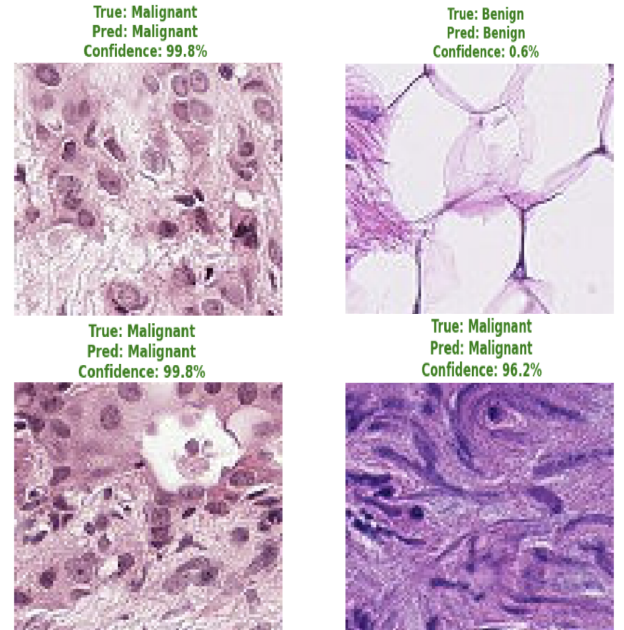


Figure 2: *Test performance evaluation.*

4. Conclusion

This work demonstrates the hybrid ResNet-ViT architectures represent a meaningful step forward in histopathological image classification, combining local feature strengths of CNNs and global contextual reasoning of Vision Transformers. This architectural family is not only effective but indispensable in CAD.

A key finding of this study is that competitive performance can be achieved without resorting to computationally heavy-weight backbones such as ConvNeXt-Base or Swin Transformers as in [8]. The proposed lightweight adaption under constrained hardware, attained a ROC-AUC of 0.9791, applying the core principles of BDCG and prototype-anchored classification to more compact solutions. This has practical implications for deployment in resource-limited settings where large-scale transformer are infeasible.

The evolution of the experiments further illustrates that pre-processing decisions are specific to each dataset. Although CLaHE demonstrated clear benefits in [15], it did not lead to performance gains for this dataset. This emphasizes the need to test preprocessing pipelines empirically, rather than assuming they can be reliably transferred between different domains.

Future work should explore decision threshold optimization, which the accuracy results suggest may not be optimal for this dataset. Additionally, incorporating learned image denoising components, such as the approach proposed by [19], may further enhance robustness to acquisition-related artifacts. Beyond this, expanding the model toward deeper residual stacks or employing hierarchical transformer encoders represents a natural avenue for pushing the performance ceiling established in this work.

5. Generative AI Use Disclosure

Gemini and Claude were used for technical adaptability, grammatical polishing, and rephrasing assistance during the preparation of this manuscript. Also for generating Figure 1.

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